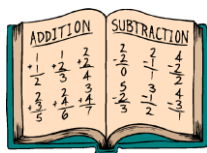


MATH REVIEW



Rules of Algebra

Rule	Example
1. $x^a x^b = x^{a+b}$	$3^2 3^4 = 3^6$
2. $(x^a)^b = x^{ab}$	$(2^3)^2 = 2^6 = 64$
3. $\sqrt{xy} = \sqrt{x}\sqrt{y}$	$\sqrt{(9)(4)} = \sqrt{9}\sqrt{4} = (3)(2) = 6$
4. $\ln(ab) = \ln(a) + \ln(b)$	$\ln(2 \times 3) = \ln(2) + \ln(3)$
5. $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$	$\ln\left(\frac{8}{3}\right) = \ln(8) - \ln(3)$
6. $\ln(a^b) = b \ln(a)$	$\ln(2^3) = 3 \ln(2)$
7. $\ln(e^x) = x$	$\ln(e^{1.4}) = 1.4$

Comment: By convention, $a^0 = 1$.

Summation Notation

In statistics, we take sums all the time. The symbol $\sum$ means “take the sum of.” When we write $\sum_{i=1}^n x_i$ it means to add n values of x :

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$$

For example, if $x_1 = 3$, $x_2 = 5$, $x_3 = -4$, and $x_4 = 2$, the sum is

$$\sum_{i=1}^4 x_i = x_1 + x_2 + x_3 + x_4 = 3 + 5 + (-4) + 2 = 6$$

In statistics, we often take sums of squared values. For example:

$$\sum_{i=1}^4 x_i^2 = x_1^2 + x_2^2 + x_3^2 + x_4^2 = 3^2 + 5^2 + (-4)^2 + 2^2 = 9 + 25 + 16 + 4 = 54$$

The sum of squares is *not* the same as the square of the sum. That is:

$$\sum_{i=1}^n x_i^2 \neq \left(\sum_{i=1}^n x_i \right)^2$$

In the previous example, the sum of squares is $\sum_{i=1}^n x_i^2 = 54$ while the square of the sum is

$$\left(\sum_{i=1}^n x_i \right)^2 = 6^2 = 36.$$

In statistics, we often need to take the sum of a product of two variables:

$$\sum_{i=1}^n x_i y_i = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

For example, if

$$x_1 = 3, x_2 = 5, x_3 = -4, x_4 = 2$$

$$y_1 = 4, y_2 = 1, y_3 = -5, y_4 = 3$$

then

$$\sum_{i=1}^4 x_i = x_1 + x_2 + x_3 + x_4 = 3 + 5 + (-4) + 2 = 6$$

$$\sum_{i=1}^4 y_i = y_1 + y_2 + y_3 + y_4 = 4 + 1 + (-5) + 3 = 3$$

$$\begin{aligned} \sum_{i=1}^4 x_i y_i &= x_1 y_1 + x_2 y_2 + x_3 y_3 + x_4 y_4 \\ &= (3)(4) + (5)(1) + (-4)(-5) + (2)(3) \\ &= 12 + 5 + 20 + 6 \\ &= 43 \end{aligned}$$

Note that the sum of the products is not the product of the sums. That is,

$$\sum_{i=1}^n x_i y_i \neq \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)$$

In the previous example, we have $\left(\sum_{i=1}^4 x_i \right) \left(\sum_{i=1}^4 y_i \right) = (6)(3) = 18$ which is not equal to

$$\sum_{i=1}^4 x_i y_i = 43.$$

Comment: Excel's function `=SUMSQ(Array)` calculate a sum of squares for an array, and its function `=SUMPRODUCT(Array1,Array2)` will calculate a sum of cross products.

Rules of Summation

$$\text{Rule 1: } \sum_{i=1}^n (x_i + y_i) = \sum_{i=1}^n x_i + \sum_{i=1}^n y_i$$

$$\text{Rule 2: } \sum_{i=1}^n (x_i - y_i) = \sum_{i=1}^n x_i - \sum_{i=1}^n y_i$$

$$\text{Rule 3: } \sum_{i=1}^n cx_i = c \sum_{i=1}^n x_i \text{ where } c \text{ is any constant}$$

$$\text{Example: } \sum_{i=1}^3 5x_i = 5x_1 + 5x_2 + 5x_3 = 5(x_1 + x_2 + x_3) = 5 \sum_{i=1}^3 x_i$$

$$\text{Rule 4: } \sum_{i=1}^n c = nc \text{ where } c \text{ is any constant}$$

$$\text{Example: } \sum_{i=1}^3 5 = 5 + 5 + 5 = (3)(5) = 15$$

Illustration: Properties of the Mean

Applying these rules, we can prove that the differences of individual observations from the mean sum to zero. The mean is defined as:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

By Rule 2:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x}$$

Since $\bar{x}$ is the same for all i , we can apply Rule 4 to get:

$$\sum_{i=1}^n \bar{x} = n\bar{x}$$

so that, using the definition of the mean $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ we get

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} = \sum_{i=1}^n x_i - n\bar{x} = \sum_{i=1}^n x_i - \cancel{n} \frac{\sum_{i=1}^n x_i}{\cancel{n}} = \sum_{i=1}^n x_i - \sum_{i=1}^n x_i = 0$$

Rules of Counting

Expression	Example
1. $n! = n(n-1)(n-2)\dots(1)$	$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$
2. ${}_nC_r = \frac{n!}{r!(n-r)!}$	${}_7C_4 = \frac{7!}{3!4!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(3 \cdot 2 \cdot 1)(4 \cdot 3 \cdot 2 \cdot 1)} = \frac{5040}{(6)(24)} = 35$

Comment 1: The expression $n!$ is called “ n factorial.” By convention, $0! = 1$.

Comment 2: The expression ${}_nC_r$ is the number of possible combinations of n things taken r at a time when order is not important. For example, if you flip 10 coins, how many different ways can you get 4 heads? There are 210 possibilities (too many to enumerate). For example: HHHHTTTTTT, HHHHTTTTTT, ... , etc. But you can easily calculate *how many* possible combinations there are:

$${}_{10}C_4 = \frac{10!}{4!(10-4)!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(4 \cdot 3 \cdot 2 \cdot 1)(6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)} = \frac{3,628,800}{(24)(720)} = 210$$

Comment 3: Cancelling terms in the numerator and denominator would reduce the magnitude of the numbers. For example, in the preceding calculation:

$${}_{10}C_4 = \frac{10!}{4!(10-4)!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot \cancel{6} \cdot \cancel{5} \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(4 \cdot 3 \cdot 2 \cdot 1)(\cancel{6} \cdot \cancel{5} \cdot 4 \cdot 3 \cdot 2 \cdot 1)} = \frac{5,040}{(24)(1)} = 210$$

Comment 4: Many inexpensive calculators have functions labeled $n!$ and ${}_nC_r$ that make these calculations easy. But if the numbers are large, the calculator will use exponential notation, which may be confusing. For example:

Example	Your calculator screen:
${}_{40}C_8 = 76,904,685$	76904685
${}_{80}C_{19} = 1.159120047 \times 10^{18}$	1.159120047 ¹⁸

EXERCISES



Algebra Exercises

Use rules for exponents to complete each expression:

E1. $3^2 3^3 =$

E2. $(4^3)^2 =$

E3. $\sqrt{(25)(16)} =$

E4. $\ln(4 \times 5) =$

E5. $\ln\left(\frac{7}{4}\right) =$

E6. $\ln(3^4) =$

E7. $\ln(e^{2.7}) =$

E8. $\ln(e^{-2.4}) =$

Combination Exercises

Evaluate each expression:

E9. $7! =$

E10. $5!6! =$

E11. ${}_6C_2 =$

E12. ${}_8C_5 =$

Summation Exercises

Given $x_1 = 4$, $x_2 = 3$, $x_3 = 7$, $x_4 = -2$ and $y_1 = 2$, $y_2 = 1$, $y_3 = 5$, $y_4 = -3$ find the following:

E13. $\sum_{i=1}^n x_i$

E14. $\sum_{i=1}^n y_i$

E15. $\sum_{i=1}^n x_i^2$

E16. $\sum_{i=1}^n y_i^2$

E17. $\sum_{i=1}^n x_i y_i$

E18. $\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)$

E19. $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

E20. $\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$

E21. $\sum_{i=1}^n (x_i + y_i)$

E22. $\sum_{i=1}^n (x_i - y_i)$

SOLUTIONS



Solutions to Algebra Exercises

Use rules for exponents to complete each expression:

$$\text{E1. } 3^2 3^3 = 3^5 = 243$$

$$\text{E2. } (4^3)^2 = 4^6 = 4096$$

$$\text{E3. } \sqrt{(25)(16)} = \sqrt{25}\sqrt{16} = (5)(4) = 20$$

$$\text{E4. } \ln(4 \times 5) = \ln(4) + \ln(5) = 1.386 + 1.609$$

$$\text{E5. } \ln\left(\frac{7}{4}\right) = \ln(7) - \ln(4) = 1.946 - 1.386$$

$$\text{E6. } \ln(3^4) = 4\ln(3) = 4(1.0986) = 4.3944$$

$$\text{E7. } \ln(e^{2.7}) = 2.7$$

$$\text{E8. } \ln(e^{-2.4}) = -2.4$$

Solutions to Combination Exercises

Evaluate each expression:

$$\text{E9. } 7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5,040$$

$$\text{E10. } 5!6! = (120)(720) = 86,400$$

$$\text{E11. } {}_6C_2 = \frac{6!}{2!4!} = 15$$

$$\text{E12. } {}_8C_5 = \frac{8!}{5!3!} = 56$$

Solutions to Summation Exercises

Given $x_1 = 4$, $x_2 = 3$, $x_3 = 7$, $x_4 = -2$ and $y_1 = 2$, $y_2 = 1$, $y_3 = 5$, $y_4 = -3$ find the following:

$$\text{E13. } \sum_{i=1}^n x_i = 12$$

$$\text{E14. } \sum_{i=1}^n y_i = 5$$

$$\text{E15. } \sum_{i=1}^n x_i^2 = 78$$

$$\text{E16. } \sum_{i=1}^n y_i^2 = 39$$

$$\text{E17. } \sum_{i=1}^n x_i y_i = 52$$

$$\text{E18. } \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) = 60$$

$$\text{E19. } \bar{x} = \frac{\sum_{i=1}^n x_i}{n} = 3$$

$$\text{E20. } \bar{y} = \frac{\sum_{i=1}^n y_i}{n} = 1.25$$

$$\text{E21. } \sum_{i=1}^n (x_i + y_i) = 17$$

$$\text{E22. } \sum_{i=1}^n (x_i - y_i) = 7$$